

H.T.No:

--	--	--	--	--	--	--	--	--	--

 **USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY: NG**
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

II B.Tech I Semester Regular Examinations, November-2024, Regulation: UR23

23BS301D: DISCRETE MATHEMATICS AND GRAPH THEORY

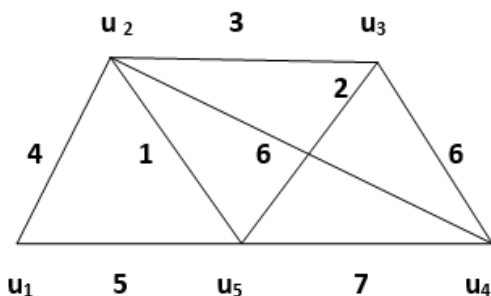
(CSE/IT/AI&DS/AI&ML)

Time: 3 hours

Max. Marks:

70

PART-A Compulsory Questions (Each carry 2 Marks)			10 X 2 =20M		
I.	a)	Define well-formed formula	CO2		(BL3)
	b)	Define Demorgan's law	CO2		(BL3)
	c)	Define equivalence relation.	CO2		(BL4)
	d)	Define cyclic permutation.	CO2		(BL4)
	e)	In how many different ways can a committee of 5 teachers and 4 students be selected from 9 teachers and 15 students?	CO3		(BL6)
	f)	Write Fibonacci relation and sequence.	CO3		(BL6)
	g)	Define order and size of a graph.	CO2		(BL3)
	h)	Define connected and disconnected graphs.	CO2		(BL3)
	i)	State Euler's formula.	CO1		(BL1)
	j)	Define complete Binary tree.	CO1		(BL1)
PART-B (Answer either A or B from each Question)			5 X 10 =50M		
II.A.	1)	Prove that for any three propositions p, q, r $\{[p \vee (q \vee r)]\} \wedge \neg q \Rightarrow (p \vee r)$	CO2	5M	(BL3)
	2)	Find the PDNF and PCNF of the formula $p \vee [\neg p \rightarrow \{q \vee (\neg q \rightarrow r)\}]$.	CO2	5M	(BL3)
OR					
II.B.	1)	Show that the premises $a \rightarrow (b \rightarrow c)$, $d \rightarrow (b \wedge \neg c)$, $a \wedge d$ are inconsistent.	CO2	5M	(BL3)
	2)	Test the validity of the following arguments Sonia is watching TV. If Sonia is watching TV, then she is not studying. If she is not studying, then her father will not buy her a scooty. Therefore, Sonia's father will not buy a scooty	CO2	5M	(BL3)
III.A.	1)	Let $A = \{1, 2, 3, 4, 6, 8, 12\}$ On A, define the partial ordering relation R by aRb if and only if $a b$. (i) Draw the Hasse diagram for ii) Write down the relation matrix for R.	CO3	5M	(BL4)
	2)	Let $A = B = \{a, b, c, d\}$ and $R = \{(a,a), (a,c), (b,c), (c,a), (d,b), (d,d)\}$ and $S = \{(a,b), (b,c), (c,a), (c,b), (d,c)\}$ compute M_R and M_S also Draw the digraph and find RoS , SoR .	CO3	5M	(BL4)
OR					

III.B.	1)	Let $A = \{1, 2, 3, 4, 6\}$ and R be a relation on A defined by aRb if and only if ' a ' is a multiple of ' b '. Represent the relation R as a matrix and draw its digraph.	CO3	5M	(BL4)
	2)	Define Lattice. Also write the properties of lattices.	CO3	5M	(BL4)
IV.A.	1)	Find the number of Permutations of the letters of the word MASSASAUGA . In how many ways of these all A's are together? How many of them begin with S?	CO4	5M	(BL6)
	2)	How many committees of 5 members with a given chair person can be selected from the 12 persons.	CO4	5M	(BL6)
		OR			
IV.B.	1)	Find the generating functions for the following sequence i) $0, 1, -2, 3, -4, \dots$ ii) $0, 2, 6, 12, 20, 30, 42 \dots$	CO5	5M	(BL6)
	2)	Solve the R.R $a_n - 7a_{n-1} + 10a_{n-2} = 4^n, n \geq 2, a_0 = 8, a_1 = 36$.	CO5	5M	(BL6)
V.A.	1)	Explain adjacency matrix and incidence matrix for a graph with suitable example.	CO4	5M	(BL3)
	2)	If G is a simple graph in which every vertex has degree at least k , prove that G contains a path of length at least k	CO4	5M	(BL3)
		OR			
V.B.	1)	Define isomorphism of two graphs. Explain with examples.	CO5	10M	(BL3)
VI.A.	1)	Prove that C_6 is a bipartite graph	CO5	5M	(BL1)
	2)	How many vertices and how many edges are there in Complete Bipartite graphs $K_{4,7}$ and $K_{7,11}$	CO6	5M	(BL1)
		OR			
VI.B.	1)	Find the minimal spanning tree for the following graph using Prim's algorithm 	CO6	10M	(BL1)

H.T.No:

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY : NG
 (An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
II B.Tech I Semester Regular Examinations, November-2024, Regulation: UR23
23HS302: UNIVERSAL HUMAN VALUES – UNDERSTANDING
HARMONY AND ETHICAL HUMAN CONDUCT
(COMMON TO ALL BRANCHES)

Time: 3 hours
70

Max. Marks:

		PART-A Compulsory Questions (Each carry 2 Marks)	10 X 2 =20M		
I.	a)	Write three benefits of Value Education.	CO1	(BL2)	
	b)	Write 5 desires and categorize them into Right Understanding, Right Relationship and Physical Facility.	CO3	(BL5)	
	c)	What are the Responses of Self and Body?	CO3	(BL5)	
	d)	What is the program for nurturing and maintaining health of the body?	CO4	(BL3)	
	e)	Explain over evaluation, under evaluation and otherwise evaluation.	CO3	(BL5)	
	f)	What are the dimensions in the society?	CO1	(BL2)	
	g)	What is the inheritance of four orders in nature?	CO1	(BL2)	
	h)	Write about Existence.	CO4	(BL4)	
	i)	How do you implement value based living in individual level?	CO1	(BL3)	
	j)	Write two criteria for Technologies.	CO6	(BL2)	
		PART-B (Answer either A or B from each Question)	5 X 10 =50M		
II.A.	1)	Make a list of your 20 desires. Now for each item on the list, find out Right Understanding, Right Relationship and Physical Facility.	CO3	10M	(BL6)

		OR			
II.B.	1)	What is happiness? Give your own experience.	CO3	10M	(BL3)
III.A.	1)	What are the needs of the self and the body?	CO2	10M	(BL2)
.					
		OR			
III.B.	1)	What are the programs for ensuring the health of the body? Explain.	CO2	10M	(BL4)
.					
IV.A.	1)	Give two examples each of over evaluation, under evaluation and otherwise evaluation.	CO4	10M	(BL4)
.					
		OR			
IV.B.	1)	What is the meaning of education and sanskāra? How does sanskāra follow education?	CO4	10M	(BL5)
.					
V.A.	1)	What do you mean by mutual fulfillment in nature? Cite a few examples.	CO5	10M	(BL1)
		OR			
V.B.	1)	Suggest ways to enhance the fulfilment of human order with the other three orders. Mention any two programs you can undertake in light of the above.	CO5	10M	(BL5)
VI.A.	1)	How does right understanding provide the basis for ethical human conduct? Give two examples.	CO5	10M	(BL5)
.					
		OR			
VI.B.	1)	How does right understanding provide the basis for humanistic constitution? Suggest some aspects of such a constitution to promote holistic living.	CO6	10M	(BL3)

H.T.No:

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY : NG
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
II B.Tech I Semester Regular Examinations, November-2024, Regulation: UR23
23ES303D: DIGITAL LOGIC & COMPUTER ORGANIZATION
(CSE/IT)

Time: 3 hours

Max. Marks: 70

		PART-A Compulsory Questions (Each carry 2 Marks)	10 X 2 =20M		
I.	a)	Convert the $(7562.45)_8$ to Binary	CO1		(BL1)
	b)	What is a MUX. Draw a logic circuit of 8 X 1 multiplexer	CO2		(BL1)
	c)	How many FFs are needed for a mod 32 counter	CO2		(BL3)
	d)	Draw the logic symbol of negative edge triggered JK FF.	CO3		(BL5)
	e)	Represent +49 in 2's complement representation	CO1		(BL3)
	f)	Give advantages of hardwired control	CO6		(BL2)
	g)	Give the main difference between RAM and ROM	CO6		(BL2)
	h)	What is volatile and non-volatile memory	CO5		(BL2)
	i)	How I/O devices are interface to processor	CO4		(BL5)
	j)	List out the various I/O interface devices	CO4		(BL3)
		PART-B (Answer either A or B from each Question)	5 X 10 =50M		
II.A.	1)	What is a MUX. Draw a logic circuit of 8 X 1 multiplexer	CO2	10M	(BL4)
		OR			
II.B.	1)	Using 2's complement perform subtraction (1101-1010)	CO2	5M	(BL1)
	2)	What are the drawbacks of half adder circuit	CO2	5M	(BL3)
III.A.	1)	What are the differences between Sequential and Combinational logic circuits	CO3	10M	(BL3)
		OR			
III.B.	1)	What is a counter and design a 4-bit Ripple up counter	CO3	10M	(BL5)
IV.A.	1)	Explain the addition and subtraction of signed magnitude numbers with examples with a flow chart.	CO5	10M	(BL1)
		OR			
IV.B.	1)	How floating-point number are represented in computers	CO5	5M	(BL3)
	2)	Explain how two floating-point are multiplied with an example	CO5	5M	(BL2)
V.A.	1)	A processor has 16 address lines and how much memory can it address and give its address map	CO5	10M	(BL4)

		OR			
V.B.	1)	What is a virtual memory and explain the techniques of implementing it	CO5	10M	(BL2)
VI.A.	1)	Explain about IO BUS and Interface modules	CO6	10M	(BL3)
		OR			
VI .B.	1)	Compare and contrast various data transfer schemes used in computer system	CO6	10M	(BL2)

H.T.No:

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY : NG
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
II B.Tech I Semester Regular Examinations, November-2024, Regulation: UR23
23PCCS304: DATABASE MANAGEMENT SYSTEM
(CSE/AI&DS/AI&ML)

Time: 3 hours
70

Max. Marks:

		PART-A Compulsory Questions (Each carry 2 Marks)	10 X 2 =20M	
I.	a)	Define the terms i) Entity set ii) Relationship set	CO1	(BL1)
	b)	Define data model.	CO1	(BL1)
	c)	What are the parts of SQL language?	CO2	(BL2)
	d)	List the set operations of SQL.	CO2	(BL2)
	e)	Write syntax on create command with example.	CO2	(BL2)
	f)	What is sub query	CO2	(BL2)
	g)	What meant by trivial dependency?	CO3	(BL1)
	h)	Define third normal form.	CO3	(BL1)
	i)	What are called secondary indices?	CO4	(BL2)
	j)	What is known as clustering file organization?	CO4	(BL2)
		PART-B (Answer either A or B from each Question)	5 X 10 =50M	
II.A.	1)	Mention various groups of database users. Explain about their roles in Detail.	CO1	10M (BL4)
		OR		
II.B.	1)	Describe in detail about one-tier, two-tier and three-tier client-server architectures	CO1	10M (BL1)
III.A.	1)	Describe all operators of relational algebra.	CO2	10M (BL1)
		OR		
III.B.	1)	Write short notes on i) DDL ii) DML iii) DCL	CO2	10M (BL4)
IV.A.	1)	Illustrate Group by and Having clauses with examples?	CO3	10M (BL1)
		OR		
IV.B.	1)	What is SQL single row function? By means of suitable examples. Illustrate the usage of SQL date, character and number functions.	CO4	10M (BL3)
V.A.	1)	Explain briefly about 1NF, 2NF with suitable examples?	CO4	10M (BL4)
		OR		

V.B.	1)	Explain about Boyce Codd normal form (BCNF) with an example.	CO5	10M	(BL4)
VI.A.	1)	Define transaction and explain desirable ACID properties of transactions	CO6	10M	(BL1)
		OR			
VI.B.	1)	Describe about the view serializability and conflict serializability	CO6	10M	(BL1)

H.T.No:

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY : NG
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

II B.Tech I Semester Regular Examinations, November-2024, Regulation: UR23
23PCCS305: OBJECT ORIENTED PROGRAMMING THROUGH JAVA
(CSE/IT/AI&DS/AI&ML)

Time: 3 hours
70

Max. Marks:

		PART-A Compulsory Questions (Each carry 2 Marks)	10 X 2 =20M	
I.	a)	Java is Platform Independent language. Justify	CO1	(BL5)
	b)	What is the difference between /(division) and % (modulus) operators.	CO1	(BL2)
	c)	What is meant by private access specifier?	CO2	(BL1)
	d)	What are the different types of constructors?	CO2	(BL1)
	e)	What is a super class?	CO2	(BL1)
	f)	Define abstract class?	CO4	(BL1)
	g)	What is the difference between error and exception in java?	CO3	(BL2)
	h)	What is Array Index Out Of Bounds Exception?	CO3	(BL2)
	i)	Give the difference between process and thread.	CO2	(BL2)
	j)	Categorize any two methods to search a string in a String Class	CO2	(BL4)
		PART-B (Answer either A or B from each Question)	5 X 10 =50M	
II.A.	1)	Demonstrate Bitwise operators using a Java program.	CO2	5M (BL3)
	2)	Develop a program to compute factorial of a number and explain the working	CO2	5M (BL6)
OR				
II.B.	1)	Explain in detail about jump statements in java	CO2	6M (BL2)
	2)	Demonstrate logical operators with an example Java program	CO2	4M (BL3)
III.A.	1)	Discuss about object oriented programming principles	CO1	10M (BL2)
OR				
III.B.	1)	Analyze about static variables and methods in detail.	CO1	10M (BL4)
IV.A.	1)	Develop a Java program to perform Binary search.	CO4	10M (BL6)
OR				
IV.B.	1)	Write java program to demonstrate the single inheritance.	CO4	5M (BL2)
	2)	Develop a Java program to implement multilevel inheritance	CO4	5M (BL6)
V.A.	1)	Explain any three classes with method in Java.util package.	CO5	10M (BL1)
OR				
V.B.	1)	List and explain the try and catch blocks in Exception handling with examples.	CO5	5M (BL1)

	2)	Explain why to use finally block in java exception handling	CO5	5M	(BL1)
VI.A.	1)	List and explain any five methods of a String class.	CO6	10M	(BL1)
		OR			
VI .B.	1)	What is batch processing? What are the required methods in batch processing?	CO6	10M	(BL1)

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY: NG
 (An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
II B.Tech II Semester Regular Examinations, April-2025, Regulation: UR23
23HM401A: MANAGERIAL ECONOMICS & FINANCIAL ANALYSIS
 (EEE/ ECE/ CSE/ IT/ AI&DS/ AI&ML)

Time: 3 hours**Max. Marks: 70**

		PART-A Compulsory Questions (Each carry 2 Marks)	10 X 2 =20M		
I.	a)	Define Income elasticity of demand?	CO1	2M	(BL1)
	b)	What is meant by Macroeconomics?	CO1	2M	(BL1)
	c)	What are the variable factors used in production?	CO2	2M	(BL1)
	d)	Define break even analysis	CO2	2M	(BL1)
	e)	What is meant by business organization?	CO3	2M	(BL1)
	f)	Define imperfect market?	CO3	2M	(BL1)
	g)	How do you estimate the working capital	CO4	2M	(BL1)
	h)	What meant by payback period?	CO4	2M	(BL1)
	i)	What is meant by double entry book keeping?	CO5	2M	(BL1)
	j)	What is meant by balance sheet?	CO6	2M	(BL1)
		PART-B (Answer either A or B from each Question)	5 X 10 =50M		
II.A.	1)	Illustrate the Law of Demand with assumptions and exceptions with the help of Individual demand schedule and demand curve.	CO2	10M	(BL2)
		OR			
II.B.	1)	How managerial economics is related to other subjects?	CO2	10M	(BL1)
III.A.	1)	Explain the law of returns to scale in long run production?	CO2	10M	(BL2)
		OR			
III.B.	1)	Explain the advantages of cost analysis?	CO3	10M	(BL2)
IV.A.	1)	Explain the features of monopolistic competition?	CO4	5M	(BL2)
	2)	Explain the different types of partners?	CO4	5 M	(BL2)
		OR			

IV.B.	1)	Explain the pricing strategies with examples?	CO4	10M	(BL2)
V.A.	1)	Explain traditional methods of capital budgeting?	CO5	5M	(BL2)
	2)	Explain the Modern methods of capital?	CO5	5M	(BL2)
		OR			
V.B.	1)	Explain the Sources of Capital budgeting?	CO5	10M	(BL2)
VI.A.	1)	Explain the importance of journal book in accounting?	CO6	5M	(BL2)
	2)	Explain the importance of double entry system?	CO6	5M	(BL2)
		OR			
VI .B.	1)	Explain the Activity ratio Ratio with formula?	CO6	5M	(BL2)
	2)	Explain the Capital Structure Ratio?	CO6	5M	(BL2)

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY: NG
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
II B.Tech II Semester Regular Examinations, April-2025, Regulation: UR23
23ES402C: PROBABILITY & STATISTICS
(Common to CSE, IT)

Time: 3 hours

Max. Marks: 70

		PART-A Compulsory Questions (Each carry 2 Marks)	10 X 2 =20M																		
I.	a)	What are the merits demerits of arithmetic mean.	CO1	(BL2)																	
	b)	Define Skewness	CO1	(BL2)																	
	c)	Write Normal equations for Parabola	CO2	(BL4)																	
	d)	Define Regression	CO2	(BL4)																	
	e)	State Baye's Theorem	CO3	(BL3)																	
	f)	Define Normal distribution	CO3	(BL3)																	
	g)	If mean $\bar{x} = 31.45$, $\mu = 32.3$, $\sigma = 1.6$ & $n = 40$ find z - test Statistic	CO6	(BL6)																	
	h)	Define Point estimator	CO6	(BL6)																	
	i)	Define Null Hypothesis	CO5	(BL4)																	
	j)	Explain the term two Tailed test	CO5	(BL4)																	
		PART-B (Answer either A or B from each Question)	5 X 10 =50M																		
II.A.	1)	Calculate the median for the following frequency distribution:	CO1	5M	(BL2)																
		<table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>f</td><td>8</td><td>10</td><td>11</td><td>16</td><td>20</td><td>25</td><td>15</td><td>9</td><td>6</td></tr></table>				x	1	2	3	4	5	6	7	8	9	f	8	10	11	16	20
x	1	2	3	4	5	6	7	8	9												
f	8	10	11	16	20	25	15	9	6												
	2)	Calculate Quartile deviation (Q.D.) from mean for the following data:	CO1	5M	(BL2)																
		<table><tr><td>Marks</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td><td>60 – 70</td></tr><tr><td>No. of students</td><td>6</td><td>5</td><td>8</td><td>15</td><td>7</td><td>6</td><td>3</td></tr></table>				Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	No. of students	6	5	8	15	7	6	3
Marks	0 – 10	10 – 20				20 – 30	30 – 40	40 – 50	50 – 60	60 – 70											
No. of students	6	5	8	15	7	6	3														
OR																					
II.B.	1)	Compute the geometric mean from the following data	CO1	5M	(BL2)																
		<table><tr><td>Marks</td><td>0 – 10</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td></tr><tr><td>No. of students</td><td>10</td><td>5</td><td>8</td><td>7</td><td>20</td></tr></table>				Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	No. of students	10	5	8	7	20				
Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50																
No. of students	10	5	8	7	20																
	2)	An incomplete frequency distribution demand as table	CO1	5M	(BL2)																
		<table><tr><td>Variables</td><td>10 – 20</td><td>20 – 30</td><td>30 – 40</td><td>40 – 50</td><td>50 – 60</td><td>60 – 70</td><td>70 – 80</td></tr><tr><td>Frequency</td><td>12</td><td>30</td><td>....</td><td>65</td><td>.....</td><td>25</td><td>18</td></tr></table>				Variables	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80	Frequency	12	30		65		25	18
Variables	10 – 20	20 – 30				30 – 40	40 – 50	50 – 60	60 – 70	70 – 80											
Frequency	12	30		65		25	18														
		Given that median is 46 and N=229.Determine the missing frequencies using the median formula.																			
III.A	1)	By the method of least squares, find the straight line that best fit to the following data	CO2	5M	(BL4)																
		<table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Y</td><td>2</td><td>5</td><td>4</td><td>7</td><td>10</td></tr></table>				X	1	2	3	4	5	Y	2	5	4	7	10				
X	1	2	3	4	5																
Y	2	5	4	7	10																
	2)	The ranks of same 16 students in Mathematics and Physics are as follows. Two numbers within brackets denote the ranks of the students in Mathematics and Physics: (1,1) (2,10) (3,3) (4,4)	CO2	5M	(BL4)																

		(5,5) (6,7) (7,2) (8,6) (9,8) (10,11) (11,15) (12,9) (13,14) (14,12) (15,16) (16,13).Calculate the rank Correlation Coefficient for proficiencies of this group in Mathematics and Physics																					
		OR																					
III.B	1)	Fit a curve of the form $y = ab^x$	CO2	5M	(BL4)																		
		<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>y</td><td>10</td><td>21</td><td>35</td><td>59</td><td>92</td><td>200</td><td>400</td><td>610</td></tr></table>	x	0	1	2	3	4	5	6	7	y	10	21	35	59	92	200	400	610			
x	0	1	2	3	4	5	6	7															
y	10	21	35	59	92	200	400	610															
	2)	Find the most likely Price in Mumbai corresponding to the Price of 70 at Kolkata from the following <table><tr><td></td><td>Kolkata</td><td>Mumbai</td></tr><tr><td>Average Price</td><td>65</td><td>67</td></tr><tr><td>S. D</td><td>2.5</td><td>3.5</td></tr></table> Correlation coefficient between Prices of commodities in the two Cities is 0.8		Kolkata	Mumbai	Average Price	65	67	S. D	2.5	3.5	CO2	5M	(BL4)									
	Kolkata	Mumbai																					
Average Price	65	67																					
S. D	2.5	3.5																					
IV.A	1)	For the continuous probability function $f(x) = kx^2e^{-x}, x \geq 0$. Find (i) k , (ii) Mean, (iii)Variance iv) SD	CO3	10 M	(BL3)																		
		OR																					
IV.B.	1)	An irregular six-faced die is thrown and the expectation that in 10 throws it will give five even numbers is twice the expectation that it will give four even numbers. How many items in 10,000 sets of 10 throws each, would you expect it to give no even number?	CO3	5M	(BL3)																		
	2)	In a distribution exactly normal, 10.03% of the items are under 25 kilogram weight and 89.97% of the items are under 70 kilogram weigh. What are the mean and standard deviations of the distribution?	CO3	5M	(BL3)																		
V.A.	1)	Samples of size 2 are taken from the population 1, 2, 3, 4, 5, 6 with replacement . Find (i) Population mean (ii)Population standard deviation (iii) Mean of the sampling distribution of means (iv) Standard deviation of the sampling distribution of mean	CO	10 M	(BL6)																		
		OR																					
V.B.	1)	A random sample of size 64 is taken from a normal population with mean $\mu=51.4$ and $\sigma=6.8$. What is the probability that the mean of the sample will i) Exceed 52.9 Fall between 50.5 and 52.3	CO6	5M	(BL6)																		
	2)	A normal population has a mean of 0.1 and SD of 2.1. Find the probability that mean of a sample of size 900 will be negative.	CO6	5M	(BL6)																		
VI.A	1)	It is claimed that a random sample of 49 tyres has a mean life of 15200 km. This sample was drawn from a population whose mean is 15150 km and standard deviation of 1200 km. Test the significance at 0.05 level.	CO5	5M	(BL4)																		
	2)	Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same against that they are not, at 5% level.	CO5	5M	(BL4)																		
		OR																					

VI .B.	1)	The heights of 10 males of a give locality are found to be 70,67,62,68,61,68,70,64,64,66 inches. Is it reasonable to believe that the average height is greater than 64 inches?	CO	5M	(BL4)
	2)	A Sample analysis of examination results of 500 students was made. It was found that 220 students had failed, 170 had secured a third class, 90 were placed in second class and 20 got a first class. Do these figures commensurate with the general examination result which is in the ratio 4 : 3 : 2 : 1 for the various categories respectively.	CO	5M	(BL4)

**USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY : NG**

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

II B.Tech II Semester Regular Examinations, April-2025, Regulation: UR23**23PCCS403: DATA WAREHOUSING AND DATA MINING****(CSE/ AI&DS/ AI&ML)****Time: 3 hours****Max. Marks: 70**

		PART-A Compulsory Questions (Each carry 2 Marks)		10 X 2 =20M
I.	a)	What is a data ware house.	CO1	(BL 1)
	b)	Define virtual warehouse.	CO1	(BL 1)
	c)	List the types of data ?	CO2	(BL 2)
	d)	Define Data quality	CO2	(BL 1)
	e)	What is the purpose of the holdout method in classification?	CO3	(BL 2)
	f)	Define Cross Validation.	CO3	(BL 2)
	g)	What is the Apriori principle?	CO5	(BL 2)
	h)	What is the FP-Growth algorithm?	CO5	(BL 1)
	i)	What are the two main types of clustering methods?	CO6	(BL 1)
	j)	What is the objective of the K-means algorithm?	CO6	(BL 2)
		PART-B (Answer either A or B from each Question)		5 X 10 =50M
II.A.	1)	Compare Operational Database Systems and Data Warehouses.	CO1	5M (BL 2)
	2)	Write short note on OLAP systems.	CO1	5M (BL2)
		OR		
II.B.	1)	Illustrate bitmap indexing OLAP data in Data Warehouse implementation.	CO1	5M (BL 2)
	2)	Demonstrate a fact constellation schema in Data Warehouse.	CO1	5M (BL 2)
III.A.	1)	Illustrate data mining tasks with examples.	CO2	10M (BL 1)
		OR		
III.B.	1)	Illustrate the terms types of data, attribute and measurement.	CO2	5M (BL 1)
	2)	Elaborate different types of attributes with example and operations.	CO2	5M (BL 5)
IV.A.	1)	Explain Naïve Bayes Classifier? How to use Bayes theorem for classification.	CO3	10M (BL 3)
		OR		
IV.B.	1)	How to build a decision tree? Explain briefly about Hunts algorithm.	CO3	5M (BL 3)
	2)	Illustrate Bayesian belief network classification with an example.	CO3	5M (BL 3)
V.A.	1)	Write about problem definition in association analysis.	CO4	10M (BL 2)
		OR		
V.B.	1)	Classify the compact representation of frequent items.	CO4	10M (BL 2)
VI.A.	1)	Classify Classical Partitioning Methods	CO5	10M (BL 4)
		OR		
VI .B.	1)	Justify Whether the purpose of the clustering is utility.	CO6	5M (BL 5)
	2)	What is a Cluster analysis and explain different ways of Clustering to same set of Points.	CO6	5M (BL 2)

H.T.No:

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY: NG
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
II B.Tech II Semester Regular Examinations, April-2025, Regulation: UR23
23PCCS404: OPERATING SYSTEM(CSE/ IT)

Time: 3 hours**Max. Marks: 70**

		PART-A Compulsory Questions (Each carry 2 Marks)	10 X 2 =20M		
I.	a)	Identify the difference between mainframe and desktop operating system.	CO1	(BL2)	
	b)	What is real time system?	CO1	(BL1)	
	c)	How will you calculate turn-around time?	CO2	(BL2)	
	d)	Difference between Process and Program	CO2	(BL2)	
	e)	What is a semaphore?	CO3	(BL1)	
	f)	What are the conditions under which a deadlock situation may arise?	CO3	(BL1)	
	g)	Define swapping.	CO4	(BL1)	
	h)	What is Belady’s anomaly?	CO4	(BL1)	
	i)	Define seek time and latency time.	CO5	(BL1)	
	j)	Define Equal Allocation.	CO5	(BL1)	
		PART-B (Answer either A or B from each Question)	5 X 10 =50M		
II.A.	1)	Discuss in detail the Operating System Functions.	CO1	10M	(BL2)
		OR			
II.B.	1)	Discuss with help of a neat diagram the structure of Operating System.	CO1	10M	(BL2)
III.A.	1)	Discuss about process scheduling based on queues with diagram	CO2	10M	(BL2)
		OR			
III.B.	1)	Explain and differentiate between user level and kernel level thread.	CO2	10M	(BL2)
IV.A.	1)	Describe about semaphores and its roles synchronization.	CO3	10M	(BL2)
		OR			
IV.B.	1)	Describe the Readers and writers’ problem. How to handle it?	CO3	10M	(BL2)
V.A.	1)	Given page reference string: 1,2,3,2,1,5,2,1,6,2,5,6,3,1,3,6,1,2,4,3. Assuming there are 3 frames Compare the number of page faults for FIFO and MRU page replacement algorithm.	CO4	10M	(BL4)
		OR			
V.B.	1)	Describe with examples the page replacement algorithms.	CO5	10M	(BL2)
VI.A.	1)	Describe the C-SCAN,LOOK and C-LOOK disk scheduling algorithm using the following data. The dist head is initially at position-cylinder 53.the cylinder sequence of requests is 98, 183, 37, 122, 14, 124, 65, 67. Find the total head movement.	CO6	10M	(BL6)
		OR			

VI .B.	1)	Explain file allocation methods in detail.	CO6	10M	(BL1)
-------------------	-----------	--	------------	------------	--------------

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY: NG
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

II B.Tech II Semester Regular Examinations, April 2025, Regulation: UR23

23PCCS405: SOFTWARE ENGINEERING

(CSE/ IT)

Time: 3 hours

Max. Marks: 70

		PART-A Compulsory Questions (Each carry 2 Marks)		10 X 2 =20M
I.	a)	List different software life cycle models?	CO1	(BL2)
	b)	Write the difference between system software and application software?	CO1	(BL1)
	c)	What are the Metrics for project size estimation?	CO2	(BL2)
	d)	List the users of SRS?	CO2	(BL1)
	e)	Define DFD?	CO3	(BL2)
	f)	What is agility in software development?	CO3	(BL1)
	g)	What is white-box testing?	CO5	(BL2)
	h)	What is debugging?	CO5	(BL2)
	i)	List the different types of CASE tools?.	CO6	(BL2)
	j)	List the software maintenance process models?	CO6	(BL2)
		PART-B (Answer either A or B from each Question)	5 X 10 =50M	
II.A.	1)	Summarize the evolution of Software Engineering, highlighting major milestones and advancements over time.	CO1	10M (BL2)
		OR		
II.B.	1)	Compare and contrast the Waterfall, Spiral, and Agile models of software development. Provide examples of situations where each model is suitable.	CO2	10M (BL4)
III.A.	1)	Analyze the various complexities involved in Software Project Management.	CO2	10M (BL4)
		OR		
III.B.	1)	Differentiate between Single Variable and Multivariable Heuristic Estimation Techniques and analyze their applicability using examples.	CO2	10M (BL4)
IV.A.	1)	Define cohesion in software design and illustrate various types of cohesion using suitable examples.	CO3	10M (BL3)
		OR		
IV.B.	1)	Compare and contrast function-oriented and object-oriented software design approaches using appropriate examples.	CO3	10M (BL4)
V.A.	1)	Explain the coding phase in software engineering. What are the objectives, standards, and guidelines involved?	CO5	10M (BL3)
		OR		
V.B.	1)	Discuss about Software Quality management system.	CO6	10M (BL3)

VI.A.	1)	Explain the architecture of a CASE environment and describe its major components and their roles.	CO6	10M	(BL2)
		OR			
VI .B.	1)	Describe the four common software maintenance models. Highlight how they differ in approach and application.	CO6	10M	(BL3)

H.T.No:

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
III B.Tech I Semester Supplementary Examinations, April-2025, Regulation: UR20
PCCS501: COMPUTER NETWORKS
(CSE)

Time: 3 hours

Max. Marks: 5x14 = 70

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	Discuss in brief about TCP/IP protocol Suite.	CO1	7M	(BL1)
	2)	Explain differences between Open System Interconnection reference model and TCP/IP reference model.	CO1	7M	(BL2)
		OR			
I.B.	1)	Describe the Transmission Media. What are the types of Transmission Media?	CO1	10M	(BL2)
	2)	Explain about bus topology with neat diagram.	CO1	4M	(BL2)
II.A.	1)	Explain in detail about the sliding window protocol using Selective Repeat.	CO2	7M	(BL2)
	2)	Explain in detail about the sliding window protocol using Go-Back-N.	CO2	7M	(BL1)
		OR			
II.B.	1)	Explain about the types of HDLC transfer modes?	CO2	6M	(BL2)
	2)	Explain about Point to Point protocol (PPP).	CO2	8M	(BL4)
III.A.	1)	Discuss the FDMA and TDMA.	CO2	7M	(BL2)
	2)	Explain about polling and token passing protocols in detail		7M	(BL2)
		OR			
III.B.	1)	Explain about pure ALOHA and slotted ALOHA? Mention the differences between them.	CO3	10M	(BL2)
	2)	Define Ethernet. List the different types of Ethernet and its implementations.	CO3	4M	(BL3)
IV.A.	1)	What are the differences between Static Routing Algorithm and Dynamic Routing Algorithm?	CO4	7M	(BL3)
	2)	Write about various classes of IP addresses.	CO4	7M	(BL2)
		OR			
IV.B.	1)	What are the closed loop congestion control techniques? Explain.	CO4	7M	(BL2)
	2)	Explain Distance Vector routing algorithm with an example.	CO4	7M	(BL2)
V.A.	1)	Explain the structure of UDP header format.	CO5	7M	(BL2)
	2)	Explain about flow control and error control in Transport Layer.	CO5	7M	(BL3)
		OR			
V.B.	1)	Explain about protocols in transport layer.	CO6	7M	(BL5)
	2)	Discuss about E-mail architecture.		7M	(BL3)



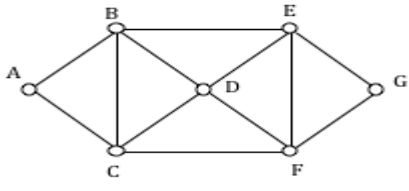
USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY
 (An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
III B.Tech I Semester Supplementary Examinations, April-2025, Regulation: UR20
PCCS502/PCIT502: DESIGN AND ANALYSIS OF ALGORITHMS
(CSE/IT)

Time: 3 hours

Max. Marks: 5x14 = 70

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	Explain Big oh, Big omega, Big teeta, Little oh notations in detail with an example?	CO1	14M	(BL4)
		OR			
I.B.	1)	Write an algorithm for factorial of a given number and find the complexities of it?	CO1	7M	(BL2)
	2)	Explain about basic efficiency classes?	CO1	7M	(BL5)
II.A.	1)	Explain about chess board problem in divide and conquer in detail with an example?	CO2	14M	(BL2)
		OR			
II.B.	1)	Explain about Dijkstra's algorithm and solve it 	CO3	14M	(BL2)
III.A.	1)	Find the all pairs shortest path from each vertex for the given adjacency matrix? $\begin{pmatrix} 0 & 1 & \infty & 1 & 5 \\ 9 & 0 & 3 & 2 & \infty \\ \infty & \infty & 0 & 4 & \infty \\ \infty & \infty & 2 & 0 & 3 \\ 3 & \infty & \infty & \infty & 0 \end{pmatrix}$	CO4	14M	(BL3)
		OR			
III.B.	1)	Explain in detail and solve with an example for multistage graphs?	CO4	7M	(BL2)
	2)	Write an algorithm for multistage graphs?	CO4	7M	(BL2)

IV.A.	1)	Explain Hamilton cycle and solve for the given graph 		14M	(BL2)
		OR			
IV.B.	1)	Explain about sum of subsets and solve $s=\{1,3,4,5\}$ and $M=8$ find the subset whose sum is 8.	CO5	14M	(BL2)
V.A.	1)	Explain in detail about the P class problems with example?	CO6	7M	(BL2)
	2)	Explain in detail about the NP class problems with example?	CO6	7M	(BL2)
		OR			
V.B.	1)	Discuss about the Node cover decision problem?	CO6	7M	(BL2)
	2)	Define Chromatic Number? And solve with an example?	CO6	7M	(BL2)

H.T.No:

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
III B.Tech I Semester Supplementary Examinations, April-2025 Regulation: UR20
PCCS503: DATA WAREHOUSING AND DATA MINING
(CSE)

Time: 3 hours
5x14 = 70

Max. Marks:

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	Explain Slice and Dice in detail with examples?	CO1	7M	(BL4)
	2)	Explain Data mining as a step in the process of knowledge discovery?	CO1	7M	(BL2)
		OR			
I.B.	1)	Different types of data warehouses and explain the components?	CO1	7M	(BL2)
	2)	Explain in detail about Star Schema and Snow flake Schema?	CO1	7M	(BL2)
II.A.	1)	Explain data transformation methods?	CO2	7M	(BL2)
	2)	Briefly explain about data cleaning techniques?	CO2	7M	(BL2)
		OR			
II.B.	1)	What is the need of preprocessing in Data mining?	CO2	7M	(BL2)
	2)	Difference between noisy data and inconsistent data?	CO2	7M	(BL2)
III.A.	1)	What are the characteristics of Decision tree induction?	CO3	7M	(BL2)
	2)	Elaborate on Navie-Bayes classifier with suitable example?	CO3	7M	(BL2)
		OR			
III.B.	1)	Explain the methods for selecting best test conditions?	CO4	7M	(BL2)
	2)	Write ashort notes on Accuracy, Error rate, Gini , Entropy and Classification Error?	CO4	7M	(BL2)
IV.A.	1)	Why we use support and confidence in association analysis?	CO5	7M	(BL2)
	2)	How to generate a candidate item set operation?	CO5	7M	(BL6)
		OR			
IV.B.	1)	Describe about the compact representation of frequent pattern item set.	CO5	14M	(BL2)
V.A.	1)	What are the main components of K-means algorithm?	CO6	7M	(BL2)
	2)	How does DB Scan determine the core points in a dataset?	CO6	7M	(BL2)
		OR			
V.B.	1)	Describe different clustering techniques?	CO6	7M	(BL2)
	2)	Differentiate between K-Means and K-Mediod clustering in terms of their underlying principles?	CO6	7M	(BL2)

H.T.No:

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
III B.Tech I Semester Supplementary Examinations, April-2025, Regulation: UR20
OEEE505A: RENEWABLE ENERGY SOURCES
(ME, CSE, IT, AI&DS, AI&ML)

Time: 3 hours
5x14 = 70

Max. Marks:

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	What are primary and secondary energy sources? Discuss in detail.	CO1	6M	(BL1)
	2)	Discuss the advantages and disadvantages of Renewable energy sources.	CO1	8M	(BL2)
		OR			
I.B.	1)	Discuss the differences between renewable and non renewable energy resources	CO1	7M	(BL2)
	2)	Write short note on energy audit	CO1	7M	(BL2)
II.A.	1)	What is solar collectors and explain different types of solar collectors?	CO2	14M	(BL2)
		OR			
II.B.	1)	What are the effects of various parameters on solar collectors?	CO2	6M	(BL1)
	2)	Discuss about i) Cylindrical parabolic concentrator ii) Fixed mirror Solar Concentrator.	CO3	8M	(BL2)
III.A.	1)	Explain the different types of solar PV systems?	CO3	14M	(BL2)
		OR			
III.B.	1)	What is the importance of Maximum power point tracking in solar PV system? Explain Perturb and Observe MPPT technique algorithm.	CO4	14M	(BL4)
IV.A.	1)	Explain different types of wind turbines with neat sketch?	CO4	14M	(BL4)
		OR			
IV.B.	1)	What is wind farm and explain different types of wind farms?	CO5	7M	(BL5)
	2)	Explain power coefficient versus interference factor characteristics with neat sketch	CO5	7M	(BL2)
V.A.	1)	Discuss the process of photosynthesis.	CO5	6M	(BL2)
	2)	List the advantages of anaerobic digestion.	CO6	8M	(BL1)
		OR			
V.B.	1)	Explain the operation of a fixed dome type biogas plant with a neat sketch.	CO	7M	(BL4)
	2)	Discuss the fuel for fuel cells in detail.	CO6	7M	(BL2)

H.T.No:

--	--	--	--	--	--	--	--	--	--



USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

III B.Tech II Semester Regular/Supplementary Examinations, April-2025, Regulation:

UR20

PCCS601/PCIT601: MACHINE LEARNING

(CSE / IT)

Time: 3 hours

Max. Marks:

5x14 = 70

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	What are the key characteristics of supervised and unsupervised learning?	CO1	7M (BL4)
	2)	Briefly discuss the concept of Statistical Learning.	CO1	7M (BL2)
		OR		
I.B.	1)	What are Risk Statistics? List and explain some commonly used risk statistics.	CO1	7M (BL2)
	2)	Explain in detail about the Tradeoffs in Statistical Learning.	CO1	7M (BL2)
II.A.	1)	Explain the process of Decision Tree construction using ID3 algorithm with an example.	CO2	14M (BL3)
		OR		
II.B.	1)	Demonstrate the importance of scoring and ranking in assessing the performance of classification tasks.	CO2	7M (BL3)
	2)	Explain attribute selection measures with example.	CO2	7M (BL2)
III.A.	1)	Discuss about Linear SVM Classification.	CO4	7M (BL2)
	2)	Discuss about Non-Linear SVM Classification.	CO4	7M (BL2)
		OR		
III.B.	1)	Explain in detail about Voting Classifiers and its types.	CO3	7M (BL2)
	2)	Explain about Stacking and Boosting.	CO3	7M (BL2)
IV.A.	1)	Define Clustering and write its applications.		7M (BL2)
	2)	Explain how to use Clustering for Image Segmentation.	CO5	7M (BL3)
		OR		
IV.B.	1)	Explain how to use Clustering for Preprocessing.	CO5	7M (BL3)
	2)	What is GMM? Discuss in detail.	CO5	7M (BL2)
V.A.	1)	Write the installation process of TensorFlow 2.	CO6	7M (BL3)
	2)	What are some common methods for Loading and Preprocessing Data with TensorFlow.	CO6	7M (BL3)
		OR		
V.B.	1)	Discuss in detail about Artificial Neural Networks with Keras.	CO6	14M (BL2)

H.T.No:

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
III B.Tech II Semester Regular/Supplementary Examinations, April-2025, Regulation:
UR20
PCCS602: COMPILER DESIGN
(CSE)

Time: 3 hours
5x14 = 70

Max. Marks:

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	Explain in detail about a) Lexical Analyzer b) Syntax Analyzer.	CO1	7M	(BL2)
	2)	Write a lex program to recognize the decimal number.	CO1	7M	(BL4)
		OR	CO1		
I.B.	1)	How many types of compilers are available? Explain about them in detail.	CO1	10M	(BL2)
	2)	Explain the terms i)Phase ii) Pass		4M	(BL2)
II.A.	1)	Explain in detail about Error recovery in Predictive Parsing	CO2	7M	(BL2)
	2)	Write the Preprocessing steps of Top-down parsing.	CO2	7M	(BL2)
II.B.	1)	What is LMD and RMD? Explain with an example.	CO2	7M	(BL2)
	2)	Consider the following grammar $S \rightarrow (L) \mid a$ $L \rightarrow L, S \mid S$ a) What are terminals and Non terminals and start symbol in the given grammar? b) Find LMD, RMD and parse tree for i) (a,a) ii) (a,(a,a)).	CO2	7M	(BL3)
III.A.	1)	Construct LALR parser for the given grammar and also parse the input string 'id=id'. $S \rightarrow L=R \mid R$ $L \rightarrow *R \mid id$ $R \rightarrow L$	CO3	14M	(BL6)
		OR			
III.B.	1)	Construct SLR parse table for the following grammar. $S \rightarrow CC$ $C \rightarrow cC \mid d$ And parse the input string 'cd'.	CO3	14M	(BL6)

IV.A.1)	Give the SDT scheme for desk calculator.	CO3	7M	(BL4)	
2)	Explain the procedure for eliminating left recursion from SDTs.	CO3	7M	(BL2)	
	OR				
IV.B.1)	Explain in detail about various forms of Intermediate Code Generation with examples.	CO4	14M	(BL2)	
V.A.1)	Explain in brief about peephole optimization techniques.	CO5	14M	(BL2)	
	OR				
V.B.1)	Explain a) Static allocation Strategy b) Stack allocation Strategy	CO5	10M	(BL2)	
2)	What are the advantages of optimizing compiler?	CO5	4M	(BL1)	

H.T.No:

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
III B.Tech II Semester Regular/Supplementary Examinations, April-2025, Regulation:
UR20
PCIT603: CRYPTOGRAPHY AND NETWORK SECURITY
(IT)

Time: 3 hours
5x14 = 70

Max. Marks:

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	Discuss any two Substitution Technique and list their merits and demerits.	CO1	7M	(BL2)
	2)	Discuss in detail about the security model for network and its principles.	CO1	7M	(BL1)
		OR			
I.B.	1)	Differentiate passive attacks from active attacks.	CO1	7M	(BL2)
	2)	What is cryptography and explain about terminology of cryptography process to provide security with neat diagram.	CO1	7M	(BL1)
II.A.	1)	Discuss the design principles of block cipher technique.	CO1	7M	(BL4)
	2)	Explain Advanced Encryption standard (AES) in detail.	CO1	7M	(BL3)
		OR			
II.B.	1)	Explain about the symmetric and asymmetric key cryptography.	CO2	7M	(BL2)
	2)	Distinguish between a Feistel and a non-Feistel block cipher.	CO2	7M	(BL3)
III.A.	1)	Illustrate ElGamal Encryption and decryption algorithm.	CO2	7M	(BL2)
	2)	Explain Chinese Remainder Theorem with an example		7M	(BL3)
		OR			
III.B.	1)	Using RSA algorithm, Find $n, \phi(n)$, d and cipher text 'C' if $p=17$, $q=11$, $e=7$ $d=23$ and $M=88$.	CO3	7M	(BL4)
	2)	Explain Asymmetric key cryptography algorithm in detail.	CO3	7M	(BL3)
IV.A.	1)	Explain the compression of Secure Hash Algorithm.	CO4	7M	(BL3)
	2)	Explain about message authentication code (MAC).		7M	(BL2)
		OR			
IV.B.	1)	Compare HMAC with CMAC.	CO6	7M	(BL3)
	2)	What are the requirements of cryptography hash functions?	CO5	7M	(BL2)
V.A.	1)	Explain S/MIME protocol.	CO5	7M	(BL3)

	2)	Explain different MIME data types and list the differences between MIME and S/MIME. Explain about IP security.		7M	(BL4)
		OR			
V.B.	1)	What is PGP? Explain different packet formats of PGP.		7M	(BL2)
	2)	Explain Secure Socket Layer architecture.		7M	(BL3)

H.T.No:

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY
 (An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
III B.Tech II Semester Regular/Supplementary Examinations, April-2025, Regulation:
UR20
PECS604B/PCIT602/PCAD602: BIG DATA ANALYTICS
(CSE/ IT/ AI&DS)

Time: 3 hours
 5x14 = 70

Max. Marks:

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	Write a major challenge in large data and how it is differentiated from specify systems?	CO1	14M	(BL1)
		OR			
I.B.	1)	Categorize the steps in Analytic Process and tools?	CO1	7M	(BL2)
	2)	Write a various type of analytics process?	CO1	7M	(BL1)
II.A.	1)	Explain in detail data stream processing platforms with individual diagrams?	CO2	14M	(BL2)
		OR			
II.B.	1)	Illustrate Flajolet-Martin Algorithm, Construct stream $X = [1, 5, 10, 5, 15, 11]$ and hash function $h(x) = x \bmod 11$, Find the count of distinct and unique elements using the Flajolet-Martin Algorithm.	CO2	14M	(BL6)
III.A.	1)	Write an information about the history of Hadoop?	CO3	14M	(BL3)
		OR			
III.B.	1)	Define MapReduce and how it works on map reduce?	CO3	14M	(BL1)
IV.A.	1)	Briefly explain about Apache Hive and how to work with Hive?	CO3	7M	(BL3)
	2)	To categorize the data processing operators in pig?	CO3	7M	(BL4)
		OR			
IV.B.	1)	Explain about the various HQL operations?	CO3	7M	(BL3)
	2)	How Zookeeper works in Hadoop?	CO3	7M	(BL6)
V.A.	1)	illustrate the data Visualization & big data with examples?	CO5	7M	(BL1)
	2)	Design an Interpretation of regression coefficients?	CO6	7M	(BL6)
		OR			
V.B.	1)	List the Methods of Visualize Data?	CO5	7M	(BL1)
	2)	Write any 5 Interactive data visualizations techniques examples?	CO5	7M	(BL6)

H.T.No:

--	--	--	--	--	--	--	--	--	--

USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
III B.Tech II Semester Regular/Supplementary Examinations, April-2025, Regulation:
UR20

OEEE605B: FUNDEMENTALS OF UTILIZATION OF ELECTRICAL ENERGY
(CSE/ IT/ AI&DS/ AI&ML)

Time: 3 hours
5x14 = 70

Max. Marks:

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	With a block diagram illustrating the components and connections involved in an electric drive system	CO1	7M	(BL3)
	2)	Explain the terms (i)continuous load (ii)intermittent load and (iii) fluctuating load	CO1	7M	(BL2)
		OR			
I.B.	1)	Derive the moment of inertia of a flywheel required for load equalization ?	CO1	14M	(BL5)
II.A.	1)	What is resistance heating and explain different types of resistance heating ?	CO2	14M	(BL2)
		OR			
II.B.	1)	What is resistance welding and explain with neat diagram	CO2	7M	(BL2)
	2)	Explain different types of electric arc furnaces with neat diagram	CO2	7M	(BL2)
III.A.	1)	Explain Rousseau’s construction for determine the mean spherical candle power(MSCP)	CO3	7M	(BL2)
	2)	Define the terms illumination, light, phase angle, solid angle?	CO3	7M	(BL1)
		OR			
III.B.	1)	Explain different methods for designing the proper lighting	CO3	7M	(BL2)
	2)	A production area in a factory 60m×24m, find the number of lamps required if each lamp has a lumen output of 18000 lumens. The illumination required for a factory area is 200 lux, utilization factor is 0.4, maintenance factor is 0.75.	CO3	7M	(BL4)
IV.A.	1)	Explain different types of methods of light control ?	CO4	7M	(BL2)
	2)	Explain the differences between filament lamps and fluorescent lamp ?	CO5	7M	(BL2)
		OR			

IV.B.	1)	What is discharge lamps and explain different types of discharge lamps ?	CO5	14M	(BL2)
V.A.	1)	Explain different types of railway services?	CO6	7M	(BL2)
	2)	Explain briefly about system of track electrification ?	CO6	7M	(BL2)
		OR			
V.B.	1)	Define the terms 1.Crest speed 2.Average speed 3. Scheduled speed	CO6	6M	(BL2)
	2)	A suburban train runs with an average speed of 36 km/h between two stations 2 km apart. Values of acceleration and retardation are 1.8 km/h/s and 3.6 km/h/s. compute the maximum speed of the train assuming trapezoidal speed/time curve?	CO6	8M	(BL4)

H.T.No:

--	--	--	--	--	--	--	--	--	--



USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

IV B.Tech I Semester Advanced Supplementary Examinations, March-2025, Regulation:

UR20

PECS701A: CLOUD COMPUTING

(CSE)

Time: 3 hours

Max. Marks:

5x14 = 70

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	Explain about Computing Paradigm Distinctions.	7M	(BL2)
	2)	What do you understand by service oriented architecture? How does it support cloud computing?	7M	(BL5)
		OR		
I.B.	1)	What are performance, security and energy efficiency in cloud computing?	14M	(BL 3)
II.A.	1)	Explain about Hardware-assisted virtualization and Full virtualization.	7M	(BL2)
	2)	Explain about Address translation in paging system.	7M	(BL2)
		OR		
II.B.	1)	What are common virtualization layers? Explain them.	14M	(BL3)
III.A.	1)	Explain IaaS, PaaS, SaaS cloud service models at different service levels.	14M	(BL2)
		OR		
III.B.	1)	Write architectural design challenges.	7M	(BL3)
	2)	Explain Market-Oriented Cloud Architecture.	7M	(BL2)
IV.A.	1)	Write and elaborate cloud resource management policies.	7M	(BL3)
	2)	Write the mechanisms for the implementation of resource management policies.	7M	(BL3)
		OR		
IV.B.	1)	Write about utility based model for cloud based web services.	7M	(BL3)
	2)	What is resource bundling? Explain combinatorial auctions.	7M	(BL2)
V.A.	1)	Explain in detail about i) Read/Write coherence ii) Before-or-after Atomicity	7M	(BL2)
	2)	Write short notes on file system and database.	7M	(BL3)
		OR		
V.B.	1)	Briefly discuss the storage architecture of GFS.	14M	(BL3)

H.T.No:

--	--	--	--	--	--	--	--	--	--

 USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY
(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)
IV B.Tech I Semester Advanced Supplementary Examinations, March-2025, Regulation:
UR20
PECS703A : BLOCK-CHAIN TECHNOLOGIES
(CSE)

Time: 3 hours
5x14 = 70

Max. Marks:

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	Explain about scenarios for Block chain	7M	(BL2)
	2)	How a Crypto currency works? Explain in detail	7M	(BL2)
		OR		
I.B.	1)	Who are participants on Block chain? Explain about players in market.	7M	(BL2)
	2)	Discuss about opportunities using Block chain	7M	(BL2)
II.A.	1)	What are two common hash functions used in Block chain explain anyone with a block diagram	7M	(BL2)
	2)	What are the challenges of a private block chain?	7M	(BL4)
		OR		
II.B.	1)	How do you create a digital identity in block chain?	7M	(BL2)
	2)	What is block chain neutrality? And how it is changing the Art World	7M	(BL2)
III.A.	1)	What are the use cases of Bitcoin? Describe them briefly	7M	(BL2)
	2)	What is the criticism of Bitcoin mining?	7M	(BL2)
		OR		
III.B.	1)	What is folding coin in block chain?	7M	(BL2)
	2)	How is a Bitcoin Script written?	7M	(BL5)
IV.A.	1)	What Are the Real-World Use Cases of Ethereum? Explain	7M	(BL2)
	2)	Why do banks reject crypto? Give a brief note	7M	(BL2)
		OR		
IV.B.	1)	How will Crypto currency affect banks?	7M	(BL2)
	2)	Discuss about Ethereum Tool Stack and Ethereum Virtual Machine. Give their merits and demerits.	7M	(BL2)
V.A.	1)	How does Bitcoin maintain anonymity?	7M	(BL2)

	2)	What happens if block chain goes down? Explain		7M	(BL2)
		OR			
V.B.	1)	Why do we need block chain governance?		7M	(BL2)
	2)	What are the applications of block chain in medical field?		7M	(BL2)

H.T.No:

--	--	--	--	--	--	--	--	--	--



USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

IV B.Tech I Semester Advanced Supplementary Examinations, March-2025, Regulation:

UR20

OEEE705B: CONCEPTS OF SMART GRID TECHNOLOGIES

(CSE)

Time: 3 hours

Max. Marks:

5x14 = 70

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	Differences between conventional grid and smart grid	CO1	7M (BL4)
	2)	Explain the Smart grid drivers and functions.	CO1	7M (BL2)
		OR		
I.B.	1)	Define smart grid and describe the Need for Smart Grid.	CO1	7M (BL2)
	2)	Explain the Concept of Resilient	CO1	7M (BL2)
II.A.	1)	What are the various types of FACTS devices incorporated into smart grids, and could you illustrate them with diagrams?	CO2	14M (BL6)
		OR		
II.B.	1)	Discuss in details about smart energy sources for smart grid function.	CO2	7M (BL1)
	2)	Explain Outage management system in detail.	CO2	7M (BL2)
III.A.	1)	Explain the physical components and layout that constitute the hardware architecture of a smart meter?	CO3	14M (BL2)
		OR		
III.B.	1)	Explain building blocks of advanced metering infrastructure	CO3	7M (BL2)
	2)	Explain challenges of AMI		7M (BL2)
IV.A.	1)	Explain the various stages of power quality audit process.	CO4	7M (BL2)
	2)	what is electromagnetic compatibility and explain the solutions for electromagnetic compatibility in smart grids	CO4	7M (BL2)
		OR		
IV.B.	1)	Explain various types of custom power devices for power mitigation	CO5	14M (BL2)
V.A.	1)	Explain broad band over power line carriers	CO5	7M (BL2)
	2)	Explain LAN and HAN	CO5	7M (BL2)
		Or		
V.B.	1)	Explain cyber security requirements in smart grid	CO6	7M (BL2)
	2)	Explain the importance of cloud computing in smart grid.	CO6	7M (BL4)

H.T.No:

--	--	--	--	--	--	--	--	--	--



USHA RAMA COLLEGE OF ENGINEERING AND TECHNOLOGY

(An Autonomous Institute with Permanent Affiliation to JNTUK, Kakinada)

IV B.Tech I Semester Advanced Supplementary Examinations, March-2025, Regulation:

UR20

OECE704A: SAFETY ENGINEERING

(CSE)

Time: 3 hours

Max. Marks:

5x14 = 70

Answer either A or B from each Question. All Questions carry equal marks.

I.A.	1)	Explain briefly different types of construction accidents?	CO1	10M	(BL2)
	2)	What is meant by accident statistics and indices?	CO1	4M	(BL1)
		OR			
I.B.	1)	Explain in detail Heinrich accident causation theory?	CO1	10M	(BL2)
	2)	Using the following data calculate the frequency rate of an accident in an construction company i. Number of workers = 500 ii. Number of last time injuries and disabilities per year =5 Average Number of hours worked by worker per year = 2000	CO1	4M	(BL4)
II.A.	1)	Define OSHA? What are the various safety standards under OSHA Act?	CO2	7M	(BL1)
	2)	a) Define safety culture? b) Define safety plan?	CO2	4M 3M	(BL1)
		OR			
II.B.	1)	What is meant by safety committee? Explain responsibilities of safety committee?	CO2	10M	(BL1)
	2)	What are the budget allocation tips?	CO2	4M	(BL1)
III.A.	1)	Define accident? Draw the accident report template?	CO3	7M	(BL1)
	2)	Define the following indices a) Incident rate b) Severity rate	CO3	7M	(BL1)
		OR	CO3		
III.B.	1)	List the types of safety training programs?	CO3	10M	(BL4)
	2)	Explain the purpose of Accident investigation?		4M	(BL2)
IV.A.	1)	Define demolition ? what are the precaution taken while doing a demolition work?	CO4	7M	(BL1)
	2)	Define concrete? What are types of concrete?	CO4	7M	(BL1)
		OR			

IV.B.	1)	What is meant by disposal of materials?	CO5	7M	(BL1)
	2)	Define temporary structure? Explain scaffolding and their types?	CO5	7M	(BL1)
V.A.	1)	What are Hazards and risks associated with the use of electricity in the workplace?	CO5	7M	(BL1)
	2)	Explain the Control measures for working near overhead power lines?	CO5	7M	(BL2)
		OR			
V.B.	1)	Explain briefly Working at height hazards and control?	CO6	14M	(BL2)